Rational Numbers and Order of Operations

Concept

Rational numbers can be written as fractions of integers with non-zero denominators.

Decimals

Terminating and repeating decimals are rational.

Signs

Subtracting means adding the opposite; sign rules differ for products.

Powers

$(-3)^2 = 9$, but $-3^2 = -9$.

Order

Brackets → powers → multiplication/division → addition/subtraction.

Example 1

Evaluate $-3/4 \div 1/2 + 2/3$.

Solution

1. Divide first: $(-3/4)(2) = -3/2$.

2. Add: $-9/6 + 4/6 = -5/6$.

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Example 2

Evaluate $1 - 2(3/5)^2$.

Solution

1. Square: $(3/5)^2 = 9/25$.

2. Multiply and subtract: $1 - 18/25 = 7/25$.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Order $-2/3$, -0.7 and $-3/5$ from least to greatest.

2. Evaluate $-5/6 + 3/4$.

3. Evaluate $(2 - 5/2)^2$.

4. Evaluate $-4 + 12 \div (-3)$.

5. Explain why $0.272727\ldots$ is rational.

6. Evaluate $[3/4 - 1/2] \div (-1/8)$.

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Practice answers | check after trying

1. $-0.7 < -2/3 < -3/5$, using $-0.700\ldots$, $-0.666\ldots$ and $-0.600\ldots$.

2. $-10/12 + 9/12 = -1/12$.

3. $(-1/2)^2 = 1/4$.

4. $-4 - 4 = -8$.

5. Let $x = 0.272727\ldots$. Then $100x - x = 27$, so $x = 27/99 = 3/11$.

6. $(1/4) \div (-1/8) = -2$. Check: $(-2)(-1/8) = 1/4$.